suggested revisions to our everyday sense of reality that are as dramatic, as mind-bending, and as paradigm-shaking as the most imaginative science fiction. These revolutionary upheavals will frame our passage through the pages that follow.

Many of the questions we explore are the same ones that, in various guises, furrowed the brows of Aristotle, Galileo, Newton, Einstein, and countless others through the ages. And because this book seeks to convey science in the making, we follow these questions as they've been declared answered by one generation, overturned by their successors, and refined and reinterpreted by scientists in the centuries that followed.

For example, on the perplexing question of whether completely empty space is, like a blank canvas, a real entity or merely an abstract idea, we follow the pendulum of scientific opinion as it swings between Isaac Newton's seventeenth-century declaration that space is real, Ernst Mach's conclusion in the nineteenth century that it isn't, and Einstein's twentieth-century dramatic reformulation of the question itself, in which he merged space and time, and largely refuted Mach. We then encounter subsequent discoveries that transformed the question once again by redefining the meaning of "empty," envisioning that space is unavoidably suffused with what are called quantum fields and possibly a diffuse uniform energy called a cosmological constant—modern echoes of the old and discredited notion of a space-filling aether. What's more, we then describe how upcoming space-based experiments may confirm particular features of Mach's conclusions that happen to agree with Einstein's general relativity, illustrating well the fascinating and tangled web of scientific development.

In our own era we encounter inflationary cosmology's gratifying insights into time's arrow, string theory's rich assortment of extra spatial dimensions, M-theory's radical suggestion that the space we inhabit may be but a sliver floating in a grander cosmos, and the current wild speculation that the universe we see may be nothing more than a cosmic hologram. We don't yet know if the more recent of these theoretical proposals are right. But outrageous as they sound, we investigate them thoroughly because they are where our dogged search for the deepest laws of the universe leads. Not only can a strange and unfamiliar reality